Getting started with Amazon Elastic Compute Cloud (Amazon EC2)

AWS Prescriptive Guidance

Amazon Elastic Compute Cloud (Amazon EC2) provides secure, resizable compute capacity in the cloud. This guide covers EC2 instance types, purchasing options, Auto Scaling, networking, storage, high availability patterns, security groups, and operational best practices. Whether you are lifting and shifting existing workloads, building cloud-native applications, or running HPC simulations, this guide will help you make the right EC2 decisions.

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Date: February 2026

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Introduction

Amazon EC2 is the core compute service of AWS, enabling you to run virtual servers — called instances — in the cloud. EC2 eliminates the need to invest in hardware upfront and allows you to develop and deploy applications faster. You can use Amazon EC2 to launch as many or as few virtual servers as you need, configure security and networking, and manage storage.

Amazon EC2 provides a broad range of instance types that are optimized for different use cases, including general-purpose, compute-optimized, memory-optimized, storage-optimized, and accelerated computing instances. Instance types comprise varying combinations of CPU, memory, storage, and networking capacity, giving you the flexibility to choose the right mix of resources for your applications.

> **Note:** EC2 instances run within a Virtual Private Cloud (VPC), providing network isolation and fine-grained control over inbound and outbound traffic using security groups and network ACLs.

Instance Types and Families

Amazon EC2 provides a large selection of instance types optimized to fit different use cases. Instance types are organized into families based on their primary use case.

Family	Examples	Primary Use Cases	Key Characteristics
General Purpose	t3, t4g, m6i, m7g	Web servers, dev/test, small databases	Balanced CPU/memory/network
Compute Optimized	c6i, c7g, c6gn	Batch processing, HPC, gaming servers	High CPU-to-memory ratio
Memory Optimized	r6i, r7g, x2idn, u-6tb1	SAP HANA, in-memory databases, big data	Large RAM capacity
Storage Optimized	i3, i4i, d3, h1	OLTP, NoSQL, distributed file systems	High sequential I/O, local NVMe
Accelerated Computing	p4, g5, inf2, trn1	ML training/inference, graphics, HPC	GPUs, FPGAs, custom ASICs
HPC Optimized	hpc6a, hpc6id	Tightly coupled HPC simulations	High-bandwidth networking, EFA

Graviton-Based Instances

AWS Graviton processors are custom-built by AWS using 64-bit Arm Neoverse technology. Graviton3-based instances (C7g, M7g, R7g) deliver up to 25% better compute performance compared to equivalent x86-based instances, with up to 50% better performance per watt. Graviton instances are an excellent choice for scale-out workloads, microservices, containerized applications, and any workload that can be compiled for the Arm architecture.

Purchasing Options

Amazon EC2 provides multiple purchasing options so you can optimize your compute costs based on your workload requirements. Choosing the right purchasing option is one of the most effective ways to reduce EC2 costs.

On-Demand Instances

On-Demand instances let you pay for compute capacity by the hour or second (minimum 60 seconds) with no long-term commitments or upfront payments. They are ideal for short-term, spiky, or unpredictable workloads that cannot be interrupted, and for applications in development or testing.

Reserved Instances (RIs) and Savings Plans

Reserved Instances and Savings Plans provide a significant discount (up to 72%) compared to On-Demand pricing in exchange for a commitment to a consistent amount of usage for a one-year or three-year term. Savings Plans are more flexible than RIs — Compute Savings Plans apply to any EC2 instance family, size, Region, operating system, or tenancy, as well as to AWS Fargate and Lambda.

Spot Instances

EC2 Spot Instances let you take advantage of unused EC2 capacity in the AWS Cloud at up to 90% discount compared to On-Demand prices. The tradeoff is that AWS can reclaim Spot capacity with a two-minute warning when it needs the capacity back. Spot Instances are ideal for fault-tolerant, stateless, and flexible workloads such as big data analytics, containerized workloads, CI/CD pipelines, image rendering, and machine learning training.

Amazon Machine Images (AMIs)

An Amazon Machine Image (AMI) is a template that contains the software configuration (operating system, application server, and applications) required to launch an instance. You can launch one or many instances from a single AMI. AMIs can be based on Amazon Linux, Ubuntu, Windows Server, Red Hat Enterprise Linux, and many other operating systems. AWS Marketplace offers thousands of commercial and community AMIs.

Custom AMIs enable a golden image strategy where you bake in your application code, configuration, and security hardening. This reduces instance startup time and ensures consistency across your fleet. You can automate AMI creation with EC2 Image Builder, which provides a managed pipeline for creating, testing, and distributing AMIs.

Networking and Security Groups

Each EC2 instance is launched into a Virtual Private Cloud (VPC) subnet. You assign one or more security groups to each instance to control inbound and outbound traffic. Security groups are stateful — return traffic is automatically allowed regardless of any rules. You can also use network ACLs at the subnet level for stateless packet filtering.

Elastic IP addresses are static public IPv4 addresses associated with your AWS account. Unlike instance public IP addresses, Elastic IPs remain associated with your account until you explicitly release them, providing a persistent endpoint for applications.

Enhanced Networking and Elastic Fabric Adapter

Enhanced Networking uses single-root I/O virtualization (SR-IOV) to provide higher bandwidth, higher packet-per-second (PPS) performance, and consistently lower inter-instance latencies. Enhanced Networking is available at no additional charge on supported instance types. For tightly coupled HPC applications, Elastic Fabric Adapter (EFA) provides OS-bypass capabilities to deliver low-latency, high-bandwidth machine-to-machine communication.

Storage Options

EC2 instances can use several storage options depending on performance, durability, and cost requirements.

Storage Type	Latency	Durability	Persistence	Typical Use Case
Amazon EBS gp3	Single-digit ms	99.8–99.9%	Persistent	Boot volumes, databases
Amazon EBS io2	Sub-ms	99.999%	Persistent	Mission-critical I/O
Amazon EBS st1	Single-digit ms	99.8–99.9%	Persistent	Throughput-intensive workloads
Instance Store (NVMe)	Microseconds	Not durable	Ephemeral	Buffers, caches, temp data
Amazon EFS	Low-to-mid ms	99.999999999 %	Persistent	Shared file system, containers
Amazon FSx	Sub-ms to ms	Highly durable	Persistent	Windows, Lustre, NetApp workloads

Auto Scaling

Amazon EC2 Auto Scaling helps you maintain application availability and allows you to automatically add or remove EC2 instances according to conditions you define. Auto Scaling uses launch templates to define instance configuration. An Auto Scaling group is a logical grouping of EC2 instances that Auto Scaling manages.

Auto Scaling supports three scaling types. Dynamic scaling responds to changing demand in real time using target tracking, step scaling, or simple scaling policies. Predictive scaling uses machine learning to schedule the right number of EC2 instances in anticipation of traffic changes. Scheduled scaling allows you to set your own scaling schedule based on predictable load changes, such as business hours or weekly spikes.

Best Practices

• Right-size your instances: Use AWS Compute Optimizer to get machine learning-powered recommendations for optimal instance types based on your actual usage.

• Use Auto Scaling groups: Always run EC2 instances in Auto Scaling groups to improve availability and enable cost optimization.

• Prefer Savings Plans over RIs: Compute Savings Plans offer greater flexibility while providing equivalent or better savings compared to Standard Reserved Instances.

• Use Graviton where possible: Graviton3 instances offer better price-performance for most workloads and reduce your carbon footprint.

• Enable detailed monitoring: Turn on EC2 detailed monitoring to receive CloudWatch metrics at 1-minute intervals for improved observability.

• Implement instance termination protection: Enable termination protection on critical instances to prevent accidental deletion.

• Use EC2 Instance Connect or AWS Systems Manager Session Manager: Avoid opening SSH/RDP ports; use managed connectivity tools for secure instance access.

• Encrypt EBS volumes: Always encrypt EBS volumes, especially for instances handling sensitive data. Use KMS customer-managed keys for additional control.

Frequently Asked Questions

Q: What is the difference between stopping and terminating an instance?

A: Stopping an instance shuts down the OS and preserves the EBS root volume, enabling you to restart it later. Terminating an instance permanently deletes it and, by default, its root EBS volume. Instance store data is lost when an instance is stopped or terminated.

Q: How do I connect to an EC2 instance without a key pair?

A: You can use AWS Systems Manager Session Manager to start an interactive shell session without managing SSH keys or opening inbound ports. EC2 Instance Connect is another option that provides temporary SSH key generation.

Q: Can I change the instance type of a running instance?

A: Yes, but you must first stop the instance. After it is stopped, you can change its instance type through the AWS Console, CLI, or API, then restart it. EBS-backed instances support instance type changes; instance store-backed instances do not.